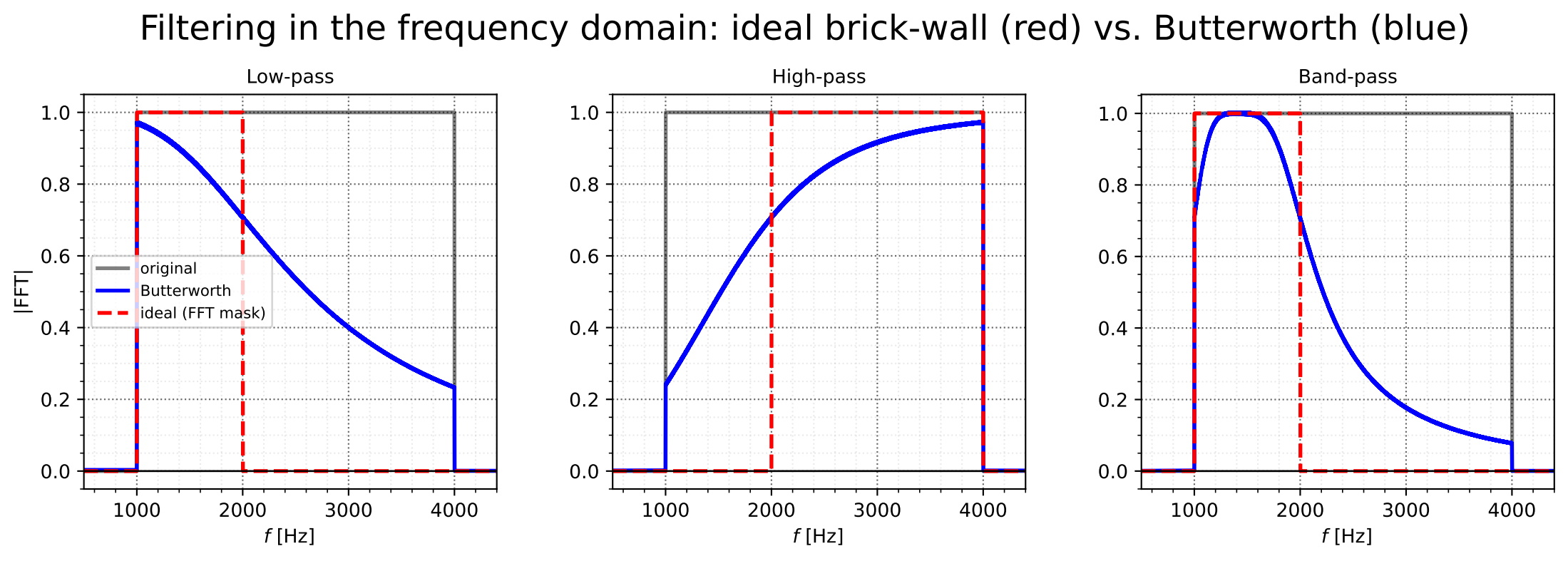

Filtering in the frequency domain: ideal brick-wall (red) vs. Butterworth (blue)
Low-pass
High-pass
Band-pass
|FFT|
f [Hz]
original
Butterworth
ideal (FFT mask)
0.0
0.2
0.4
0.6
0.8
1.0
1000
2000
3000
4000